

Mike Roe Mouse: Parts, Specifications, and Schematics

Clayton Freedman, Dorian Trotman, Labid Hossain,
Matt Lechocinski, Imran Sayeed

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1 Electronics Parts

1.1 STM32F405RGT6 Microcontroller

The F-series STM32 microcontroller will handle all computational tasks. This includes navigation and maze-solving algorithms, IR emitter control, and vacuum fan control.

1.1.1 Purchasing Information

As of July 9th, 2026

Name	Per Unit	Quantity	Total	Vendor	Link
STM32F405RGT6	\$5.85	1	\$5.85	JLC	Link

1.1.2 Dimensions

10 mm x 10 mm

1.1.3 Electrical Specifications

I/O pins CMOS and TTL compliant.

Not all I/O pins are 5V compliant.

Logic level voltage values from p.110 of data sheet.

Logic Level	Logic Type	Min	Max
V_{IH}	CMOS	$0.7V_{DD}$	5.0V
V_{IL}	TTL	0V	0.8V
V_{IH}	TTL	2.0V	5.0V

Maximum voltage and current ratings for V_{DD} and I/O.

V_{DD} ranges are with respect to ground indicated by V_{SS} .

V_{DD} ranges and I/O ranges on pp. 76–77 of data sheet.

Max V_{DD}	Min V_{DD}	Max $I_{V_{DD}}$ Sink	Max $I_{V_{SS}}$ Source
4.0V	-0.3V	150mA	150mA

Max $I_{I/O}$ Sink	Max $I_{I/O}$ Source
25mA	25mA

1.2 SFH4545 Infrared Emitter

Infrared emitter that the micromouse uses to detect distance from walls.

Distance calculated based on return time for infrared emitter.

1.2.1 Purchasing Information

As of July 17th, 2026

Name	Price Per Unit	Quantity	Total Cost	Vendor	Item Link
SFH4545	\$0.79	4	\$3.16	DigiKey	Link

1.2.2 Electrical Specifications

Operating current and voltage requirements. On pp. 4–5 of spec sheet.

Max Constant Forward Current	Max Pulsed Forward Current
100mA	1A

Max Constant Forward Voltage	Max Pulsed Forward Voltage
1.7V	2.9V

Operational centroid wavelengths

Minimum	Typical	Maximum
920mm	940mm	960mm